Adaptive Planning for Multi-Agent Workflows

An empirical study of manager agents: Handling Preference, Goal and Environmental Changes in Multi-Agent Workflows

Research Direction

- Use an established multi-agent benchmark and add a layer of mid-workflow changes.
- Run a baseline manager agent, analyse its errors for each change type, then test targeted improvements one at a time.
- Compare against reference conditions (ignore the change, replan from scratch), not against other published methods.

Problem Definition

In multi-agent workflows, a manager agent decomposes a task, delegates it to worker agents and tracks progress. During execution the workflow can change after some work is already committed: the environment changes (for example road closures or delays), the user's preferences shift (how options are ranked), or the user's goals change (what counts as success). The manager must decide which completed and in-progress work to keep, which to revise, and how to reassign the remainder. Restarting wastes completed work, while ignoring the change delivers the wrong outcome.
This study evaluates how a baseline manager agent handles each change type, analyses its failures, and tests which training-free improvements help for which type.

Research Questions

1. How does a baseline manager agent behave under environmental, preference and goal changes, and what are its main failure modes?
2. Which training-free design factors reduce those failures, and for which change type? Candidates are an explicit continue, repair or replan decision, identifying affected tasks from dependencies, and tracking the active requirements.
3. How do results depend on when the change arrives and how much of the remaining plan it affects?
4. What does each improvement cost in model calls?

Why This Is a Real Problem

- Changes after work starts are normal. Requirements change in real projects and cause delays and extra cost (study), and users of AI agents add and revise requests as the agent works (PersonaForge).
- Agents handle changes poorly. Performance drops when the user changes what they want (Tack et al.), and strong models struggle when interrupted mid-task (InterruptBench).
- Restarting is costly. Repairing an existing plan usually needs fewer changes than replanning from scratch (Fox et al., 2006), and enterprise systems already handle requests that change while in progress (Oracle).
- Many agents make it harder. Planners fail to adapt when they do not track what is already done (REALM-Bench), and many multi-agent failures come from unclear or lost instructions (MAST). How often users change their minds in multi-agent workflows has not been measured, so this is stated as an assumption.

Setting

- Benchmark: MA-Gym existing benchmarks
- Environmental changes: the benchmark's existing disruptions (traffic, road closures, delays).
- Preference and goal changes: injected at controlled points. A preference change alters how options are ranked. A goal change alters what counts as success.
- Reference answer: where an exact solver exists, re-solve the remaining problem under the new objective.

Method

1. Baseline: define and freeze one manager agent (model, prompt, visible state, delegation).
2. Error analysis: categorise failures for each change type, for example ignoring the update, restarting everything, losing track of completed work, or replanning the wrong part.
3. Improvements: for each frequent failure, add one targeted modification, test it alone, then in combination.
4. Held-out check: develop on one set of scenarios and report final results on a separate set.

Variables

- Change type: environmental, preference, goal.
- Timing: early, mid or late in the workflow.
- Share of the remaining plan affected by the change.

- Whether the change is stated explicitly or only implied.
- One factor is varied at a time from a default setting.

What kind of Benchmark is MA-Gym

Capability	How MA-Gym covers it
Planning	Actions to decompose a task, refine it and add dependencies
Delegation	Actions to assign one task or all pending tasks. The Assign-All baseline gets higher goal completion but worse constraint adherence
Coordination	Dependencies gate execution, so human actions can block downstream tasks (a CoT run averaged 46.9 simulated hours)
Monitoring	Actions for workflow status, task inspection and available agents
Communication	A messaging action, scored by a stakeholder-management metric
Adaptation	Partial. Preference change-points and workers joining or leaving mid-run
Governance	Hard and soft constraints feed the score, but the paper treats the wider topic mostly as a research agenda

Evaluation

- Outcome quality after the change, compared with the re-solved reference.
- Unnecessary changes to work the change does not affect, using unaffected tasks as a control.
- Cost in model calls and tokens.
- Repeated runs for each condition to report variability.

Scope and Limitations

- One benchmark and cheap model tiers. Task changes and agent unavailability are future work.
- instances, so it is used as a controlled testbed for LLM managers.
- Findings from a planning benchmark may not transfer to other workflows.

Next Steps

1. Understand the problems with ma-gym benchmarks
2. Understand what I can add from other peer-reviewed benchmarks
3. Will inject
    a. No changes
    b. Environment changes
    c. Goal changes
    d. Preference changes
4. Run experiments starting with baseline which is just manager agent with CoT
5. Then understand the metrics
6. Empirical study making improvements to manager agent with CoT